

Lateral Meningocele Syndrome (Lehman Syndrome): A Comprehensive Disease Characterization

Disease: Lateral Meningocele Syndrome (LMS) **Synonyms:** Lehman syndrome **Identifiers:** OMIM %130720 · MONDO:0007537 · ORPHA:2140 · Gene: NOTCH3 (HGNC:7883), chr19p13.12
Category: Mendelian, autosomal dominant

Summary

Lateral Meningocele Syndrome (LMS), also known as Lehman syndrome, is an extremely rare autosomal-dominant multisystem connective-tissue and skeletal disorder caused by **heterozygous truncating variants in the last coding exon (exon 33) of the *NOTCH3* gene**. These variants delete the C-terminal PEST degradation domain, stabilizing the intracellular NOTCH3 fragment and producing a **gain-of-function** in Notch signaling. This mechanism sharply distinguishes LMS from CADASIL — the common cerebral small-vessel disease that is allelic to LMS but caused by cysteine-altering missense variants in the NOTCH3 extracellular domain acting through a fundamentally different, aggregation-based pathway.

Clinically, LMS is defined by **multiple bilateral lateral spinal meningoceles** (the severe end of the dural ectasia spectrum, most prominent in the thoracolumbar spine) accompanied by a **distinctive craniofacial gestalt** (hypertelorism, telecanthus, high-arched eyebrows, ptosis, midfacial hypoplasia, micrognathia), **connective-tissue features** (skin hyperextensibility, joint hypermobility, hernias, scoliosis), hypotonia, and **skeletal osteopenia**. The phenotype is progressively expanding and now includes short stature, congenital heart defects, feeding difficulties/failure to thrive, developmental delay/intellectual disability, sensorineural hearing loss, inner-ear anomalies, renal cystic disease, and — most recently — biliary anomalies. Diagnosis rests on **whole-spine MRI** demonstrating the characteristic meningoceles combined with **molecular confirmation of an exon-33 *NOTCH3* variant**; prenatal molecular diagnosis is now feasible.

Mechanistic work in mouse and human iPSC models has established that the **bone-loss component is driven by NOTCH3-enhanced, RANKL-mediated osteoclastogenesis**, with NOTCH3 acting on osteoblast-lineage cells and osteocytes to upregulate RANKL. This paralog-specific effect (NOTCH1 inhibits, NOTCH2/NOTCH3 promote osteoclastogenesis) mirrors the pathobiology of Hajdu-Cheney syndrome (NOTCH2). No FDA-approved disease-modifying therapy exists; current management is supportive and neurosurgical. However, **Notch3-lowering antisense oligonucleotides (ASOs) and an anti-Notch3 neutralizing antibody reverse the skeletal phenotype in male mouse models**, and allele-selective ASOs achieve 70–80% knockdown of mutant transcript in patient-derived cells — a promising but still preclinical therapeutic avenue.

Key Findings

F001 — LMS is caused by heterozygous truncating *NOTCH3* variants clustered in the last exon (exon 33)

The causal gene was identified by exome resequencing of five unrelated individuals with LMS, all of whom carried heterozygous truncating *NOTCH3* mutations; a sixth case was confirmed by Sanger sequencing. In total, **five novel *de novo* mutations were identified in six unrelated patients**. The reported variants include a 26-bp deletion (c.6461_6486del, p.G2154fsTer78), a recurrent single-base insertion (c.6692_93insC, p.P2231fsTer11, seen in two patients), and three nonsense variants (c.6247A>T p.K2083; c.6663C>G p.Y2221; c.6732C>A p.Y2244). *A defining molecular feature is that all variants cluster in the last coding exon (exon 33)*, producing premature termination and truncation of the C-terminal region.*

"We performed exome resequencing in five unrelated individuals with LMS and identified heterozygous truncating NOTCH3 mutations." — PMID: 25394726

"All mutations cluster into the last coding exon, resulting in premature termination of the protein and truncation of..." — PMID: 25394726

Additional confirmatory variants have since been reported, including a de novo c.6723_6736del p.(Glu2241AspfsTer8) identified through the 100,000 Genomes Project ([PMID: 40256810](#)) and an 80-bp deletion in exon 33 ([PMID: 26754023](#)), reinforcing the exon-33 truncation signature. **Identifiers:** NOTCH3 (HGNC:7883); OMIM gene 600276; disease OMIM %130720.

F002 — LMS variants cause NOTCH3 gain-of-function through loss of the PEST domain

The exon-33 truncations delete the **C-terminal PEST domain**, a proline-, glutamate-, serine-, and threonine-rich sequence that normally targets the intracellular Notch fragment for proteasomal degradation. Its loss **stabilizes the NOTCH3 intracellular domain**, prolonging and enhancing downstream Notch signaling. This was directly demonstrated in human iPSC models: cells harboring the NOTCH3 c.6692_93insC variant showed **enhanced expression of the canonical Notch target genes HES1, HEY1, HEY2, and HEYL**, confirming gain-of-function.

"Lateral Meningocele or Lehman Syndrome (LMS) is associated with NOTCH3 mutations causing deletions of the PEST domain and a gain-of-NOTCH3 function." — [PMID: 35760307](#)

"NOTCH3 6692-93insC cells displayed enhanced expression of Notch target genes HES1, HEY1, 2 and L demonstrating a NOTCH3 gain-of-function." — [PMID: 39752389](#)

Functional consequence: gain-of-function via protein stabilization (not haploinsufficiency). **Suggested GO terms:** Notch signaling pathway (GO:0007219); positive regulation of transcription by RNA polymerase II (GO:0045944); protein stabilization (GO:0050821).

F003 — Bone loss/osteopenia in LMS is driven by NOTCH3-enhanced RANKL-mediated osteoclastogenesis

The skeletal osteopenia of LMS has been mechanistically dissected in mouse models. The **Notch3^{em1Ecan} knock-in mouse**, harboring a 6691TAATGA PEST-truncating mutation homologous to human LMS variants, exhibits **both cancellous and cortical bone osteopenia**. Experimental analysis revealed that the bone loss is **secondary to increased osteoclastogenesis driven by enhanced RANKL (receptor activator of NF-κB ligand)**

expression by osteoblast-lineage cells. Osteocyte-specific induction of the Lehman mutation independently caused osteopenia in male mice, localizing a key cellular source of the RANKL signal to osteocytes.

"Experimental mouse models of LMS revealed that the bone loss is secondary to increased osteoclastogenesis due to enhanced expression of receptor activator of nuclear factor kappa B ligand by cells of the osteoblast lineage." — PMID: 33519922

Suggested GO terms: osteoclast differentiation (GO:0030316); positive regulation of osteoclast differentiation (GO:0045672); bone resorption (GO:0045453). **Suggested CL terms:** osteoblast (CL:0000062); osteocyte (CL:0000137); osteoclast (CL:0000092). **Protein:** TNFSF11/RANKL.

F004 — Notch3-targeting ASOs and an anti-Notch3 antibody reverse the skeletal phenotype in LMS mouse models (preclinical)

Because stabilized gain-of-function NOTCH3 is the disease driver, **lowering NOTCH3 is a rational therapeutic strategy**. Subcutaneous administration of Notch3 ASOs (25–50 mg/kg) decreased Notch3 mRNA in liver, heart, and bone and ameliorated cortical osteopenia, reducing femoral cortical porosity in Notch3^{em1Ecan} mice. The ASOs were paralog-specific (they did not downregulate Notch1, Notch2, or Notch4). **Allele-selective ASOs targeting the mutant insertion reduced mutant transcript by 70–80% in mesenchymal cells**, offering an approach that spares the wild-type allele. Separately, a **Notch3-neutralizing antibody also reversed the skeletal phenotype in male mice** (PMID: 31188489).

"Lateral Meningocele Syndrome (LMS) is a monogenic disorder associated with NOTCH3 pathogenic variants that result in the stabilization of NOTCH3 and a gain-of-function." — PMID: 37704069

"In vivo, the subcutaneous administration of Notch3 ASOs at 25 to 50 mg/Kg decreased Notch3 mRNA in the liver, heart and bone." — PMID: 35536858

These interventions remain **preclinical**; there is no FDA-approved therapy for LMS. **Suggested NCIT terms:** Antisense Oligonucleotide; Monoclonal Antibody Therapy.

F005 — LMS is a multisystem connective-tissue/skeletal syndrome defined by lateral meningoceles plus distinctive facies

The **core diagnostic triad** comprises (1) multiple lateral thoracolumbar spinal meningoceles — the severe end of the dural ectasia spectrum, typically most severe in the lower spine; (2) a distinctive facial gestalt; and (3) connective-tissue signs.

"Facial features of LMS include hypertelorism and telecanthus, high arched eyebrows, ptosis, midfacial hypoplasia, micrognathia, high and narrow palate, low-set ears and a hypotonic appearance." — PMID: 25394726

"The characteristic lateral meningoceles represent the severe end of the dural ectasia spectrum and are typically most severe in the lower spine." — PMID: 25394726

The **expanded phenotype** includes short stature, congenital heart defects, feeding difficulties/failure to thrive, developmental delay/intellectual disability, sensorineural hearing loss, renal cysts, Chiari I malformation, syringomyelia, hydrocephalus, and tethered cord.

"Besides the lateral meningoceles, this condition presents with dysmorphic features, short stature, congenital heart defects, and feeding difficulties." — PMID: 32141180

There is substantial phenotypic overlap with Hajdu-Cheney syndrome (NOTCH2), Marfan syndrome, Ehlers-Danlos syndrome, and Loeys-Dietz syndrome, complicating clinical recognition (PMID: 25821090; PMID: 40256810).

Suggested HPO terms: Meningocele (HP:0002435); Dural ectasia (HP:0100775); Hypertelorism (HP:0000316); Telecanthus (HP:0000506); Ptosis (HP:0000508); Micrognathia (HP:0000347); Highly arched eyebrow (HP:0002553); Midface retrusion (HP:0011800); Joint hypermobility (HP:0001382); Hyperextensible skin (HP:0000974); Scoliosis (HP:0002650); Muscular hypotonia (HP:0001252); Short stature (HP:0004322); Sensorineural hearing impairment (HP:0000407); Chiari type I malformation (HP:0007099); Syringomyelia (HP:0003396); Tethered cord (HP:0002144).

F006 — LMS is a very rare, mostly de novo autosomal-dominant disorder (~few dozen reported cases)

LMS is described as "very rare" to "exceedingly rare." A 2019 neurosurgical review identified only **11 articles covering 16 cases (9 males, 7 females) across 14 families**; among those genetically screened, all carried exon-33 *NOTCH3* truncations. The original gene-discovery cohort reported **five de novo mutations in six unrelated patients**. Two instances of vertical (parent-to-child) transmission have been documented, consistent with autosomal-dominant inheritance, but most cases are de novo. Age at diagnosis ranges from infancy (as early as 5 months / 2 years) to 55 years.

"Our literature search revealed 11 articles (16 cases) of LMS, which included 9 males and 7 females, belonging to 14 different families." — [PMID: 31838470](#)

"In total, five novel de novo NOTCH3 mutations were identified in six unrelated patients." — [PMID: 25394726](#)

No formal population prevalence or incidence figures are established; Orphanet lists prevalence as <1/1,000,000 (ORPHA:2140). The near-balanced sex ratio (~9:7 M:F) is consistent with autosomal-dominant inheritance without sex bias. A very late diagnosis at age 55 illustrates that milder cases may go unrecognized for decades ([PMID: 24311540](#)).

F007 — LMS and CADASIL are distinct allelic *NOTCH3* disorders with opposite functional mechanisms

NOTCH3 is associated with two mechanistically opposite diseases. **CADASIL** — the most common monogenic cerebral small-vessel disease — is caused by **stereotyped cysteine-altering missense variants in the EGF-like repeats of the NOTCH3 extracellular domain** that change the number of cysteine residues, causing ectodomain misfolding, aggregation (granular osmiophilic material, GOM), and vascular smooth muscle cell degeneration. **LMS** is caused by **PEST-domain truncating variants** producing intracellular gain-of-function. Rare patients carrying both classes of variant have been reported.

"CADASIL, the most common monogenic form of cSVD, is caused by stereotyped mutations in the NOTCH3 receptor that alter the number of cysteine residues in its extracellular domain." — [PMID: 40145673](#)

"the patient was found to have two variants of the NOTCH3 gene, resulting in the diagnosis of lateral meningocele (Lehman) syndrome" — PMID: 34172679

Additional NOTCH3-related phenotypes further map the genotype-mechanism landscape: **biallelic loss-of-function** variants cause a neurodevelopmental disorder with spasticity and childhood-onset stroke, whereas **biallelic cysteine-involving missense** variants produce a CADASIL-spectrum phenotype (PMID: 39191170). The molecular pathobiology of CADASIL — non-enzymatic NOTCH3 fragmentation and cysteine-redox-driven aggregation — has been characterized in detail (PMID: 31901894; PMID: 35409031; PMID: 35223989), underscoring how different its mechanism is from LMS.

F008 — Diagnosis relies on whole-spine MRI plus molecular confirmation of an exon-33 NOTCH3 variant; management is neurosurgical/supportive

Whole-spine MRI is the key imaging study, demonstrating multiple bilateral well-defined cystic masses within the neural foramina (predominantly thoracolumbar), with neural foraminal widening, dural ectasia, and posterior vertebral body scalloping. CT complements this by showing vertebral/pedicle scalloping and spinal canal widening. **Molecular confirmation** is by exome/genome or targeted NOTCH3 exon-33 sequencing; "reverse phenotyping" (identifying the variant first, then imaging) has diagnosed presymptomatic and prenatal cases.

"showed multiple bilateral well-defined cystic masses within the neural foramina involving the entire spine, predominantly the thoracolumbar regions, with neural foraminal widening and dural ectasia suggestive of multiple lateral meningoceles" — PMID: 33042242

"an early genomic analysis allowed us to recognize the presence of lateral meningoceles and to begin early monitoring of her condition for possible neurological complications" — PMID: 34121137

Management is **supportive and neurosurgical**, with no formal guidelines. Symptomatic meningoceles are treated by surgical repair or shunting (cyst-subarachnoid or cystoperitoneal). A recent case documents cyst-subarachnoid shunts at T8 and L5-S1 producing immediate symptomatic improvement and gradual meningocele regression.

"Two cyst-subarachnoid (C-S) shunts were placed, at the superior aspect of the meningocele (T8) and the inferior aspect (L5-S1). His symptoms improved immediately, and the meningocele gradually regressed postoperatively." — PMID: 41432782

Surgery is complicated by coexisting Chiari I malformation, syringomyelia, hydrocephalus, tethered cord, and the underlying mesodermal/connective-tissue fragility (PMID: 31838470; PMID: 38755334).

F009 — Genetically engineered and natural mouse and iPSC models recapitulate the LMS skeletal phenotype

The principal in vivo model is the **Notch3^{em1Ecan} knock-in mouse** (6691TAATGA PEST-truncating mutation), whose heterozygotes reproduce cancellous and cortical bone osteopenia and increased femoral cortical porosity. Osteocyte-specific induction of a NOTCH3 Lehman mutation causes osteopenia in male C57BL/6J mice. **Human iPSC models** (NCRM1/NCRM5 carrying NOTCH3 c.6692_93insC plus isogenic controls) differentiated toward neural crest, mesenchymal, and osteogenic lineages exhibit gain-of-function and enhanced osteogenesis. A spontaneous murine *Notch3* mutation ("humpback") provides a natural model.

"We created a mouse model (Notch3^{em1Ecan}) harboring a 6691TAATGA mutation in the Notch3 locus, and heterozygous Notch3^{em1Ecan} mice exhibit cancellous and cortical bone osteopenia." — PMID: 35536858

"induced pluripotent NCRM1 and NCRM5 stem (iPS) cells harboring a NOTCH3 6692-93insC insertion were created" — PMID: 39752389

These models enabled therapeutic testing (ASOs and antibody rescue). **Phenotype recapitulation is strongest for the skeletal/bone-loss component**; the models less fully capture the meningocele and craniofacial features. The "humpback" natural mutant was precisely genotyped using a PCR-based ARMS system (PMID: 33860007). **Resources:** MGI (Notch3 alleles); Alliance of Genome Resources. **Orthologous gene:** mouse *Notch3* (NCBI Gene 18131).

F010 — NOTCH paralogs have opposing effects on bone; NOTCH3 specifically induces osteoblast/osteocyte RANKL

Paralog specificity explains the LMS bone phenotype. In the skeleton, **NOTCH1 inhibits osteoclastogenesis, whereas NOTCH2 enhances osteoclast differentiation, and NOTCH3 induces RANKL expression in osteoblasts and osteocytes**, thereby driving osteoclast differentiation via an indirect mechanism. This is why NOTCH3 gain-of-function in LMS produces net bone resorption/osteopenia — mechanistically parallel to NOTCH2 gain-of-function in Hajdu-Cheney syndrome.

"NOTCH3 induces the expression of RANKL in osteoblasts and osteocytes and as a result induces osteoclast differentiation." — [PMID: 32526405](#)

"NOTCH1 inhibits osteoclastogenesis, whereas NOTCH2 enhances osteoclast differentiation and function by direct and indirect mechanisms." — [PMID: 32526405](#)

"There are no effective therapies for LMS." — [PMID: 33519922](#)

F011 — The phenotype is progressively expanding; prenatal diagnosis is feasible and new organ involvements continue to be reported

The **first prenatal molecular diagnosis of LMS** was achieved by prenatal exome sequencing following an ultrasound showing fetal cystic hygroma, mild bilateral ventriculomegaly, and facial dysmorphisms; postnatal MRI confirmed lateral meningoceles and evaluation revealed **previously unreported biliary anomalies**.

"We report the first case of prenatal molecular diagnosis of LMS, which was made using prenatal exome sequencing after an ultrasound with findings of fetal cystic hygroma, mild bilateral ventriculomegaly, and facial dysmorphisms." — [PMID: 40771185](#)

"A complete clinical evaluation was performed and unexpected biliary anomalies were found. The occurrence of biliary anomalies has not been previously reported in LMS" — [PMID: 40771185](#)

Earlier reports had already expanded the spectrum to include inner-ear abnormalities and multicystic kidney disease.

"expands the spectrum of clinical manifestations related to LMS to include inner ear abnormalities and multi-cystic kidney disease" — [PMID: 32141180](#)

Strikingly, a molecularly confirmed 8-year-old case **lacked lateral meningoceles entirely**, sharing only the dysmorphic facies, G-tube dependence, failure to thrive, and developmental delay — expanding the phenotype and cautioning that the hallmark meningoceles are not obligate ([PMID: 39119451](#)). Consistent with the multisystem picture, NOTCH3 functions as a transcriptional activator across diverse tissues.

Full Section-by-Section Report

1. Disease Information

LMS is a rare hereditary connective-tissue disorder characterized by multiple lateral spinal meningoceles, distinctive facial dysmorphism, joint/skin laxity, hypotonia, and skeletal, cardiac, and urogenital anomalies. **Key identifiers:** OMIM %130720; MONDO:0007537; ORPHA:2140; MeSH — indexed under "Meningocele" (no dedicated LMS descriptor); ICD-10 Q06.8 / ICD-11 LA05.Y as nearest structural codes (no LMS-specific code). **Synonyms:** Lehman syndrome; Lateral meningocele syndrome. The information base is a mixture of **individual patient case reports/case series** (dominant, given rarity) and **aggregated disease-level resources** (OMIM, Orphanet).

2. Etiology

Primary cause: genetic — heterozygous truncating variants in the last exon (exon 33) of *NOTCH3* (F001), acting via a **gain-of-function** through PEST-domain loss and protein stabilization (F002). **Genetic risk factors:** monogenic and essentially fully explained by the *NOTCH3* exon-33 variant; no human modifier loci mapped. **Environmental risk factors:** none identified — LMS is not known to be influenced by toxins, lifestyle, or infectious exposures; most cases arise as de novo germline events. **Protective factors:** none described. **Gene-environment interactions:** no evidence — this is a highly penetrant Mendelian disorder.

3. Phenotypes

Phenotypes span physical malformations (lateral meningoceles, dural ectasia, craniofacial dysmorphism, scoliosis), clinical signs (joint hypermobility, skin hyperextensibility, hypotonia), and functional/developmental abnormalities (feeding difficulties, developmental delay/intellectual disability, sensorineural hearing loss). See F005 and F011 for the full catalog and HPO mappings. **Onset** is congenital/neonatal-to-childhood; meningoceles were historically identified before age 8 (average age of identification ~4 years), though a meningocele-free case at age 8 and a first diagnosis at age 55 demonstrate wide variability in age of recognition. **Severity** is variable, from severe infantile presentations with failure to thrive to mild adult presentations with chronic musculoskeletal pain. **Progression:** meningoceles/dural ectasia can be progressive (enlarging via CSF pulsation); the syndrome overall is chronic and lifelong. **Frequency:** lateral meningoceles and characteristic facies are near-universal (rare exceptions); connective-tissue signs, hypotonia, and feeding difficulties are common; cardiac, renal, hearing, and biliary involvement are variable. **Quality-of-life impact:** substantial — chronic pain, joint instability, neurological sequelae from meningoceles (including iatrogenic nerve damage after surgery), feeding/growth problems, developmental disability. No formal EQ-5D/SF-36 data exist for this ultra-rare disease.

4. Genetic/Molecular Information

Causal gene: *NOTCH3* (HGNC:7883; OMIM gene 600276), chr19p13.12. **Variant types:** frameshift (deletions, insertions) and nonsense variants in exon 33 — all truncating, all removing the PEST domain (F001). Representative variants: c.6461_6486del (p.G2154fsTer78); c.6692_93insC (p.P2231fsTer11); c.6247A>T (p.K2083); c.6663C>G (p.Y2221); c.6732C>A (p.Y2244); c.6723_6736del (p.E2241fsTer8); an 80-bp exon-33 deletion. *Classification: pathogenic per ACMG (PVS1-type truncating in a gene with an established truncating/GoF mechanism, de novo PS2, phenotype-specific). Allele frequency: absent from population databases (gnomAD) — private, de novo variants. Origin: germline, predominantly de novo; two documented vertical transmissions. Functional consequence: gain-of-function through protein stabilization (F002). Modifier genes: none established. Epigenetic changes: none reported. Chromosomal abnormalities:* none — LMS is a single-gene disorder.*

5. Environmental Information

No environmental, lifestyle, or infectious contributors are implicated. LMS is a fully genetic, de novo–predominant Mendelian disorder. (Note: lateral meningoceles as an isolated radiological finding can occur in neurofibromatosis type 1 and as sporadic lesions — e.g., [PMID: 15688204](#), [PMID: 23607071](#), [PMID: 38755334](#) — but these are distinct from *NOTCH3*-driven LMS.)

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

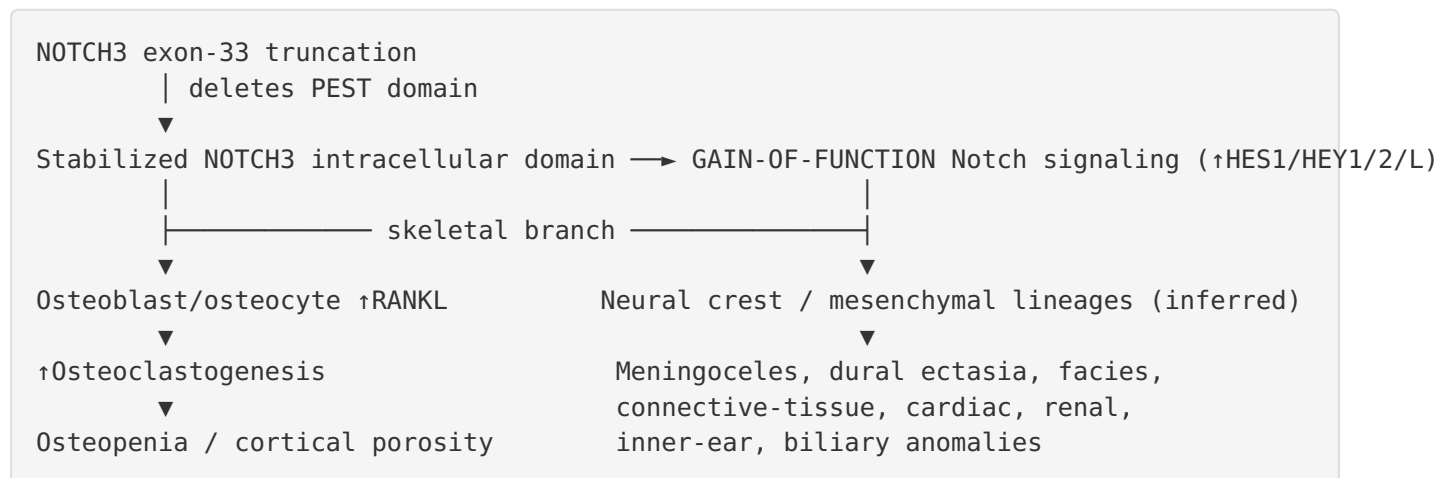
1. A **de novo heterozygous truncating variant in *NOTCH3* exon 33** arises in the germline (demonstrated).
2. The truncation **deletes the C-terminal PEST domain** of the *NOTCH3* intracellular domain (demonstrated).
3. Loss of the PEST degron **impairs proteasomal turnover of the cleaved *NOTCH3* intracellular domain, stabilizing it** (demonstrated via gain-of-function readouts; direct stabilization inferred and supported by analogy to Hajdu-Cheney *NOTCH2*).
4. Stabilized *NOTCH3* **leads to enhanced and prolonged canonical Notch signaling** — elevated HES1, HEY1, HEY2, HEYL (demonstrated in iPSC models).
5. In the skeleton, this gain-of-function **branch** proceeds: enhanced *NOTCH3* signaling in **osteoblasts and osteocytes results in increased RANKL (TNFSF11) expression** (demonstrated in mouse models).
6. Elevated RANKL **drives increased osteoclast differentiation and activity** (demonstrated), which **results in cancellous and cortical bone osteopenia and increased cortical porosity** (demonstrated).
7. In parallel developmental **branches** (mechanistically less resolved), *NOTCH3* gain-of-function in neural crest–derived and mesenchymal lineages is **inferred to lead to** the meningocele/dural ectasia, craniofacial, connective-tissue, cardiac, renal, inner-ear, and biliary phenotypes. These arms are demonstrated clinically but the intervening cellular mechanism is inferred rather than experimentally mapped.

Molecular pathway: Notch signaling (KEGG hsa04330; Reactome "Signaling by *NOTCH3*").

Cellular processes: osteoclastogenesis, bone resorption, cell-fate specification. **Protein**

dysfunction: loss of PEST-mediated degradation → stabilization → gain-of-function (contrast CADASIL's extracellular aggregation, F007). **Key cell types (CL):** osteoblast (CL:0000062), osteocyte (CL:0000137), osteoclast (CL:0000092), neural crest cell (CL:0000333, inferred), mesenchymal stem cell (CL:0000134). **Key GO processes:** Notch signaling pathway

(GO:0007219), osteoclast differentiation (GO:0030316), positive regulation of osteoclast differentiation (GO:0045672), bone resorption (GO:0045453). **Upstream vs downstream:** the *NOTCH3* variant and PEST loss are upstream; RANKL induction and osteoclastogenesis are downstream effectors of the bone phenotype.



7. Anatomical Structures Affected

Primary organs/systems: nervous system and meninges (spinal dura/arachnoid — lateral meningoceles, dural ectasia; UBERON:0002360 meninges, UBERON:0002240 spinal cord) and skeletal system (vertebrae with scalloping, generalized osteopenia; UBERON:0001474 bone). **Secondary/variable involvement:** cardiovascular (congenital heart defects; UBERON:0000948 heart), genitourinary (renal cysts; UBERON:0002113 kidney), auditory (inner ear/cochlea; UBERON:0001846), hepatobiliary (biliary anomalies; UBERON:0002394 bile duct), craniofacial skeleton and soft tissues, and skin/connective tissue (UBERON:0002097). **Tissue types:** connective tissue, nervous tissue (meninges), bone. **Cell populations:** osteoblasts, osteocytes, osteoclasts, neural crest-derived and mesenchymal cells. **Subcellular:** nucleus (GO:0005634 — NOTCH3 ICD acts as a transcriptional co-activator); plasma membrane (GO:0005886 — receptor); cytoplasm (site of stabilized fragment). **Localization/lateralization:** meningoceles are **bilateral and multiple**, predominantly **thoracolumbar**, most severe in the lower spine.

8. Temporal Development

Onset: congenital; features usually recognized in infancy or childhood. **Onset pattern:** chronic/insidious. **Progression:** chronic and lifelong; meningoceles/dural ectasia can slowly enlarge; osteopenia is progressive. **Course:** stable-to-progressive rather than episodic or

relapsing-remitting. **Critical periods:** prenatal/early-childhood windows for diagnosis and neurological monitoring; surgical timing is dictated by symptomatic meningoceles. **Duration:** lifelong.

9. Inheritance and Population

Inheritance: autosomal dominant (F006), most cases de novo with two documented vertical transmissions. **Penetrance:** high/complete for the molecular phenotype, though the meningocele feature is not fully obligate (F011). **Expressivity:** highly variable (infantile-severe to adult-mild). **Anticipation:** not described. **Germline mosaicism:** not formally documented but plausible given de novo predominance. **Founder effects/consanguinity:** not applicable (de novo dominant). **Carrier frequency:** not applicable. **Epidemiology:** prevalence <1/1,000,000 (Orphanet); only a few dozen cases reported worldwide; near-balanced sex ratio (~9:7 M:F). **Age distribution:** diagnoses span 5 months to 55 years.

10. Diagnostics

Imaging (central): whole-spine MRI showing multiple bilateral cystic neural-foraminal masses, neural foraminal widening, dural ectasia, and vertebral scalloping; CT for bony scalloping and canal widening (F008). **Genetic testing (confirmatory):** targeted *NOTCH3* exon-33 sequencing, or WES/WGS; CMA/karyotype are not informative (single-nucleotide/small-indel disorder). Reverse phenotyping (genetics-first) has diagnosed presymptomatic and prenatal cases. **Prenatal:** prenatal exome sequencing after ultrasound findings of cystic hygroma, ventriculomegaly, and facial dysmorphism (F011). **Biomarkers/labs:** no specific biochemical biomarker; no routine metabolomic/proteomic diagnostic. **Clinical criteria:** no formal consensus criteria; diagnosis is gestalt (facies + meningoceles + connective-tissue signs) plus molecular confirmation. **Differential diagnosis:** Hajdu-Cheney syndrome (NOTCH2), Marfan, Ehlers-Danlos (hypermobility/classic), Loeys-Dietz, arterial tortuosity syndrome, neurofibromatosis type 1 (isolated meningoceles), and Copenhagen syndrome (radiological mimic) — distinguished by the *NOTCH3* exon-33 variant and full multisystem gestalt ([PMID: 25821090](#); [PMID: 40256810](#)).

11. Outcome/Prognosis

Survival/mortality: no systematic survival data; LMS is generally not rapidly lethal, though severe infantile presentations with congenital heart defects and failure to thrive carry higher morbidity/mortality risk. **Morbidity:** chronic musculoskeletal pain, joint instability, neurological complications from meningoceles, developmental disability, feeding/growth

impairment. **Complications:** neurological sequelae (including iatrogenic nerve damage after meningeal surgery — PMID: 24311540), CSF hypotension from meningoceles, Chiari I/syringomyelia/hydrocephalus/tethered cord. **Recovery:** neurosurgical shunting can improve symptoms and regress meningoceles (PMID: 41432782), but the syndrome is not curable. **Prognostic factors:** severity of meningoceles/neurological involvement, cardiac anomalies, feeding difficulties. No validated prognostic biomarkers.

12. Treatment

No FDA-approved disease-modifying therapy exists (F004, F010). Current management is **supportive and neurosurgical** (F008): neurosurgical repair or cyst-subarachnoid/cystoperitoneal shunting of symptomatic meningoceles; management of scoliosis, feeding difficulties (G-tube), cardiac and renal anomalies; physical/occupational therapy for hypotonia and joint instability; audiology support for hearing loss. **Pharmacotherapy:** none disease-specific; the RANKL-axis logic suggests antiresorptives (e.g., denosumab, an anti-RANKL antibody) could be rationally explored for osteopenia, but this is not established for LMS. **Emerging/experimental (preclinical):** Notch3-lowering ASOs (including allele-selective ASOs achieving 70–80% mutant-transcript knockdown) and an anti-Notch3 neutralizing antibody reverse the skeletal phenotype in mouse and iPSC models (F004, F009). **Suggested NCIT terms:** Antisense Oligonucleotide Therapy; Monoclonal Antibody Therapy; Surgical Procedure; Cerebrospinal Fluid Shunt. **Pharmacogenomics:** not applicable.

13. Prevention

No primary prevention exists for a de novo genetic disorder. **Secondary prevention:** early molecular diagnosis (including prenatal) enables early neurological monitoring and timely surgical intervention (F008, F011). **Genetic counseling:** recurrence risk is low for parents of a de novo case but 50% for an affected individual's offspring (autosomal dominant); prenatal/preimplantation testing is feasible where a familial variant is known. **Tertiary prevention:** surveillance for and management of neurological, cardiac, renal, and orthopedic complications. No immunization or public-health/environmental interventions apply.

14. Other Species / Natural Disease

LMS is a human disorder; no naturally occurring LMS-equivalent disease is documented in companion animals or wildlife (no OMIA equivalent). The gene is evolutionarily conserved: mouse *Notch3* (NCBI Gene 18131) is orthologous to human *NOTCH3*. A **spontaneous murine *Notch3* mutation ("humpback")** exists as a natural model ([PMID: 33860007](#)). No zoonotic or cross-species transmission applies (non-infectious genetic disease).

15. Model Organisms

Mammalian (mouse): the engineered *Notch3^{em1Ecan}* knock-in (6691TAATGA PEST truncation) recapitulates cancellous and cortical osteopenia; osteocyte-specific conditional induction of the Lehman mutation causes osteopenia in male C57BL/6J mice; the spontaneous "humpback" *Notch3* mutant is a natural model (F009). **Cellular/in vitro:** human iPSC lines (NCRM1/NCRM5 with NOTCH3 c.6692_93insC plus isogenic controls) differentiated to neural crest, mesenchymal, and osteogenic cells demonstrate gain-of-function and enhanced osteogenesis (F002, F009). **Genetic model types:** knock-in, conditional (osteocyte-specific), spontaneous mutant, patient-derived iPSC. **Phenotype recapitulation:** strong for the skeletal/bone-loss component (enabling therapeutic proof-of-concept for ASOs and antibody); **limitations:** do not robustly reproduce the lateral meningoceles or full craniofacial/multisystem phenotype. **Resources:** MGI (*Notch3* alleles), Alliance of Genome Resources.

Mechanistic Model / Interpretation

LMS is best understood as a **NOTCH3 stabilization disorder**. The single molecular lesion — an exon-33 truncation that removes the PEST degron — converts NOTCH3 into a longer-lived, hyperactive transcriptional co-activator. The clearest, experimentally validated downstream consequence is in bone: gain-of-function NOTCH3 in osteoblasts and osteocytes upregulates RANKL, tipping the RANKL/OPG balance toward osteoclast activation and net resorption, producing the osteopenia captured in mouse models. The remaining multisystem features (meningoceles, dural ectasia, craniofacial dysmorphism, connective-tissue laxity, cardiac/renal/inner-ear/biliary anomalies) are clinically well-documented but mechanistically inferred to arise from NOTCH3 gain-of-function in neural-crest and mesenchymal lineages during development — a gap current models do not fully bridge.

The disease sits within an instructive **NOTCH3 allelic and paralogous framework**:

Disorder	Gene	Variant class	Domain	Mechanism	Core phenotype
LMS (Lehman)	NOTCH3	Truncating (exon 33)	Intracellular PEST	Gain-of-function (stabilization)	Lateral meningoceles, osteopenia, facies
CADASIL	NOTCH3	Cysteine-altering missense	Extracellular EGF repeats	Ectodomain aggregation (GOM)	Cerebral small-vessel disease, stroke, dementia
Biallelic LoF NOTCH3	NOTCH3	Biallelic loss-of-function	—	Loss-of-function	Neurodevelopmental disorder, spasticity, childhood stroke
Hajdu-Cheney	NOTCH2	Truncating (last exon)	Intracellular PEST	Gain-of-function	Acro-osteolysis, osteoporosis (paralog analogue)

The parallel between LMS (NOTCH3) and Hajdu-Cheney syndrome (NOTCH2) — both caused by last-exon PEST-truncating gain-of-function variants and both producing bone loss — is a powerful cross-validation of the mechanism and explains why the RANKL-osteoclast axis is central to both.

Evidence Base

PMID	Contribution	Evidence type
25394726	Landmark gene discovery: exon-33 truncating <i>NOTCH3</i> variants; facial and meningocele phenotype	Human clinical/genetic
35760307	PEST deletion → gain-of-function; osteocyte-specific mouse osteopenia	Model organism
39752389	iPSC gain-of-function (↑ HES1/HEY targets); ASO targeting	In vitro (human iPSC)
33519922	RANKL-mediated osteoclastogenesis mechanism; "no effective therapies"	Review/model organism
35536858	Notch3 ^{em1Ecan} model; in vivo ASO knockdown and osteopenia rescue	Model organism
37704069	ASO amelioration of cortical osteopenia; drug-target rationale	Model organism
31188489	Anti-Notch3 antibody reverses skeletal phenotype	Model organism
32526405	NOTCH paralog specificity; NOTCH3→RANKL→osteoclast axis	Review/mechanistic
31838470	Neurosurgical review; case count and sex ratio	Human clinical review
32141180	Phenotype expansion (inner ear, multicystic kidney)	Human clinical
40771185	First prenatal molecular diagnosis; biliary anomalies	Human clinical
39119451	LMS without lateral meningoceles — phenotype expansion	Human clinical
40145673	CADASIL mechanism (extracellular cysteine-altering) — contrast	Review
34172679	Co-occurring CADASIL/Lehman variants; craniosynostosis	Human clinical
39191170	Biallelic NOTCH3 genotype-phenotype spectrum	Human clinical
33042242	Diagnostic MRI appearance (infantile Lehman)	Human clinical
41432782	Cyst-subarachnoid shunt with meningocele regression	Human clinical

PMID	Contribution	Evidence type
34121137	Genetics-first (reverse phenotyping) diagnosis in infant	Human clinical
33860007	"Humpback" natural murine Notch3 mutant	Model organism
40256810	Novel variant; radiological mimicry (Copenhagen syndrome)	Human clinical
26754023	14th case; 80-bp exon-33 deletion; cardiac/feeding phenotype	Human clinical
24311540	Late diagnosis at 55; vertical transmission; DDx overlap	Human clinical
25821090	LMS within connective-tissue-disorder differential	Review
31901894 · 35409031 · 35223989	CADASIL pathobiology (fragmentation/aggregation) — mechanistic contrast	In vitro/review

Limitations and Knowledge Gaps

- 1. Ultra-rarity limits epidemiology.** With only a few dozen reported cases, there are no reliable prevalence/incidence, survival, or natural-history statistics; sex ratio and age-of-onset estimates derive from tiny case series.
- 2. Mechanistic gap for non-skeletal features.** The RANKL-osteoclast mechanism is firmly established for bone, but the cellular pathways linking NOTCH3 gain-of-function to meningoceles, dural ectasia, craniofacial dysmorphism, and cardiac/renal/inner-ear/biliary anomalies remain inferred, not demonstrated.
- 3. Model-organism incompleteness.** Mouse and iPSC models recapitulate the skeletal phenotype well but do not reproduce the hallmark meningoceles, limiting preclinical testing of therapies for the most clinically significant feature.
- 4. No human therapeutic data.** ASO and antibody efficacy is entirely preclinical and demonstrated primarily in male mice; sex-specific effects, meningocele response, and human safety are unknown.

5. **Phenotype still expanding.** Newly reported organ involvements (biliary) and atypical presentations (no meningoceles) indicate the full phenotypic spectrum and its frequencies are not yet defined.
 6. **No validated diagnostic criteria or biomarkers,** and no genotype-phenotype correlation study large enough to relate specific exon-33 variants to severity.
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Proposed Follow-up Experiments / Actions

1. **Develop a meningocele-capturing model** — conditional NOTCH3-Lehman induction in neural-crest/meningeal lineages (e.g., Wnt1-Cre, PDGFR β) to test whether dural ectasia and meningoceles can be reproduced and mechanistically dissected.
 2. **Test RANKL-axis pharmacology** — evaluate denosumab (anti-RANKL) or bisphosphonates in LMS mouse models and consider registry monitoring of bone outcomes in patients, given the well-defined RANKL mechanism.
 3. **Advance allele-selective ASOs toward IND** — extend the 70–80% mutant-knockdown data to systemic and CNS delivery, female cohorts, and long-term safety, with meningocele and craniofacial endpoints where possible.
 4. **Establish an international LMS registry and natural-history study** — pool cases to define phenotype frequencies, age-of-onset distributions, complication rates, surgical outcomes, and genotype-phenotype correlations.
 5. **Define a multi-organ surveillance protocol** — recommended cardiac, renal, hepatobiliary, audiology, and spinal imaging screening intervals for molecularly confirmed patients, informed by the expanding phenotype.
 6. **Deep molecular profiling of iPSC-derived neural crest/mesenchyme** — transcriptomic and proteomic characterization to identify effectors bridging NOTCH3 gain-of-function to connective-tissue and craniofacial phenotypes, and to nominate additional druggable nodes.
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Report compiled from a 5-iteration autonomous investigation: 11 confirmed findings and 30 reviewed papers. Evidence types are distinguished as human clinical/genetic, model organism, in vitro (human iPSC), and review/mechanistic throughout.

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